CHAPTER 1

Number Sequences

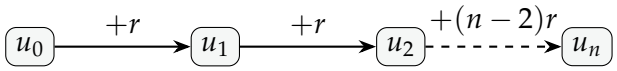
Skills: investigate, model, represent, calculate, communicate

1.1 Core Course Notes

A numerical sequence assigns to each natural number n a number u_n. The index n denotes a position in the sequence; it must not be confused with the value u_n.

1.1.1 Arithmetic sequence

> **Key point.** A sequence (u_n) is **arithmetic** when there is a real number r such that, for all n, $u_{n+1} = u_n + r$. The number r is the common difference. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.



Three numbers x, y, z, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, that is, if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of x and z.

> **Method.** To model an additive situation: 1) define u_n and state the unit; 2) identify the first term; 3) express the constant increase or decrease as the common difference r; 4) write the general term; 5) answer with a sentence in context.

> **Example 1.** A shop sells 480 units in the first month and then 25 more units each month. With $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: the shop expects to sell 755 units in the twelfth month.

1.1.2 Sum of arithmetic terms

For the $n+1$ terms from u_0 to u_n,

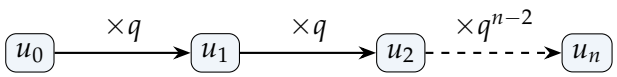
$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula is easy to remember as: *number of terms multiplied by the mean of the first and last terms*.

> **Watch out.** In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 9 or 12.

1.1.3 Geometric sequence

> **Key point.** A sequence (u_n) is **geometric** when there is a real number q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.



An increase of $t\,\%$ corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of $t\,\%$ to $q = 1 - \dfrac{t}{100}$. Three positive terms x, y, z are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean. For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

1.1.4 Algorithms: computing a sum

```
def somme_carres(n):
    s = 0
    for k in range(1, n + 1):
        s = s + k**2
    return s
```

The accumulator `s` holds the partial sum; the counter `k` runs through the integers from 1 to n.

Session 1 — Arithmetic sequences: recognising and modelling

Number Sequences

15 Sep 2026 • B: 2 h – C: 1 h

Objectives.

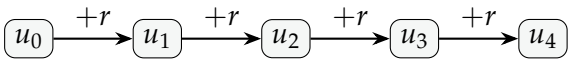
- Recognise an additive change.
- Find the common difference.
- Define a sequence in context.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. .....
2. Calculate u_4 if $u_0 = 14$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from u_3 to u_{15}? .....
4. Give the multiplier for a 8% decrease. .....
5. Give the common difference of $12, 12+4, 12+2\times 4, \ldots$. .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise an additive change and find the common difference.

Activity 1. A club has 320 members. It expects to gain 18 more each year. Let u_n be the number of members after n years. Calculate u_0, u_1, u_2. Which operation takes you from one term to the next? Write a relation between u_{n+1} and u_n, then predict u_5.



Written record. A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a real-life situation, a constant absolute increase leads to an arithmetic sequence.

Example 2. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.1 · essential.

1. The sequence (u_n) satisfies $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is worth 420 and decreases by 3% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, course notes pp. 18–21; exercises p. 24, no. 1–3. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A warehouse manager is testing three models. For every natural number n, n is the week number.

- Number of parcels prepared: $a_n = 3(2n + 7) - n$.
- Number of customer complaints: $b_n = 2n^2 - n + 4$.
- Number of pallets still in stock: $c_n = 150 - 7n$.

a. For each of the three sequences, calculate $u_{n+1} - u_n$ (with $u = a$, b or c) and state whether the sequence is arithmetic; if so, give its common difference.

b. For each arithmetic sequence, interpret the common difference in context.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 125 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 2 — **Arithmetic sequences: the general term**

Number Sequences — 17 Sep 2026 • B: 1 h – C: 2 h

Objectives.

- Move from the recurrence relation to the general term.
- Calculate a distant term.
- Interpret the index.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 5 - x + 2$. .....
2. How many terms from u_3 to u_{15}? .....
3. Give the multiplier for a 8% decrease. .....
4. Give the common difference of $12, 12 + 5, 12 + 2 \times 5, \ldots$. .....
5. Calculate u_4 if $u_0 = 15$ and $u_{n+1} = u_n + 3$. .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: move from the recurrence relation to the general term and calculate a distant term.

Activity 2. A shop has 1,250 active loyalty cards at index 0 and gains 75 per month. Without calculating all the terms, find the number of cards at index n, then at index 18. Explain the role of 18 in the calculation.

Written record. If (u_n) is arithmetic with common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the index of the first term.

Example 3. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.2 · essential.

1. The sequence (u_n) satisfies $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write u_n then calculate u_{12}.
2. A quantity is worth 440 and decreases by 4% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, course notes pp. 18–21; exercises p. 24, no. 5–8. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A delivery company is gradually replacing its diesel vehicles with electric ones. At the start of 2025 (rank 0), it owns 186 diesel vehicles and it withdraws 4 of them every year. Let d_n be the number of diesel vehicles at the start of the year $2025 + n$; the sequence (d_n) is arithmetic.

a. Give the first term d_0 and the common difference of the sequence (d_n).

b. Express d_n in terms of n for every natural number n.

c. Calculate d_{12} and d_{30}. Interpret d_{30} in a sentence.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 130 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 3 — **Sum of the terms of an arithmetic sequence**

Number Sequences — 22 Sep 2026 • B: 2 h – C: 1 h

Objectives.

- Count the terms of a sum.
- Apply the formula.
- Check the order of magnitude.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 120. .....
2. Give the multiplier for a 8% decrease. .....
3. Give the common difference of $12, 12+6, 12+2\times 6, \ldots$. .....
4. Calculate u_4 if $u_0 = 16$ and $u_{n+1} = u_n + 3$. .....
5. How many terms from u_3 to u_{15}? .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: count the terms of a sum and apply the formula.

Activity 3. A company pays a monthly bonus: 45 euros in the first month, then 5 euros more each month. Calculate the total paid over the first twelve months in three ways: explicit sum, pairing first and last terms, formula.

Written record. A sum of arithmetic terms equals number of terms $\times \dfrac{\text{first} + \text{last}}{2}$. From u_p to u_q, there are $q - p + 1$ terms.

Example 4. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.3 · essential.

1. The sequence (u_n) satisfies $u_0 = 130$ and $u_{n+1} = u_n + 5$. Write u_n then calculate u_{12}.
2. A quantity is worth 460 and decreases by 5% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, course notes pp. 20–21; exercises p. 24, no. 11–13, then p. 28, no. 43–44. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application.

a. Calculate $\sum_{i=1}^{365} i = 1 + 2 + \cdots + 365$.

b. Calculate $\sum_{k=100}^{365} k = 100 + 101 + \cdots + 365$. State the number of terms in this sum.

c. To build up some savings, Lova takes on a challenge: on the k-th day of the year, she puts k cents into a money box (1 cent on day 1, 2 cents on day 2, and so on). How much, in euros, will she have put in altogether over a 365-day year? And from day 100 to day 365 inclusive?

Homework. Redo practice exercise 2 with the following data: replace the initial value by 135 and the rate or common difference/ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 4 — Geometric sequences: recognising and modelling

Number Sequences

24 Sep 2026 • B: 1 h – C: 2 h

Objectives.

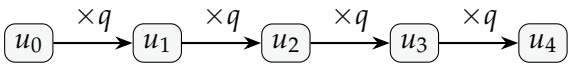
- Recognise a multiplicative change.
- Convert a rate into a multiplier.
- Compare arithmetic and geometric sequences.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 25% increase. .....
2. Give the common difference of $12, 12+7, 12+2\times 7, \ldots$. .....
3. Calculate u_4 if $u_0 = 17$ and $u_{n+1} = u_n + 3$. .....
4. How many terms from u_3 to u_{15}? .....
5. Give the multiplier for a 8% decrease. .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise a multiplicative change and convert a rate into a multiplier.

Activity 4. An online shop increases its number of visitors by 6% per month, starting from 8,000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of sequence.



Written record. A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A constant relative change leads to a geometric sequence.

Example 5. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.4 · essential.

1. The sequence (u_n) satisfies $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write u_n then calculate u_{12}.
2. A quantity is worth 480 and decreases by 6% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, course notes pp. 23–25; exercises p. 25, no. 14–18. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. An app is being tested before its launch. In week 0, 3 testers install it. We assume that, each week, the number of new testers is multiplied by 4. Let v_n be the number of new testers in week n; the sequence (v_n) is geometric.

a. Give the first term and the common ratio of the sequence (v_n), then express v_n in terms of n.

b. Calculate v_5 and v_9.

c. Does the model seem realistic over a long period? Justify in a sentence.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 140 and the rate or common difference/ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 5 — Geometric sequences: general term and sum

Number Sequences

29 Sep 2026 • B: 2 h – C: 1 h

Objectives.

- Write the general term.
- Calculate a geometric sum.
- Interpret a cumulative change.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 8 - x + 2$. .....
2. Calculate u_4 if $u_0 = 18$ and $u_{n+1} = u_n + 3$. .....
3. How many terms from u_3 to u_{15}? .....
4. Give the multiplier for a 8% decrease. .....
5. Give the common difference of $12, 12 + 8, 12 + 2 \times 8, \ldots$. .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: write the general term and calculate a geometric sum.

Activity 5. A capital of 250,000 Ar is invested at 4% per year, compound interest. Write the capital C_n after n years. Calculate C_6, then the sum $C_0 + \cdots + C_6$, and explain what this last sum represents mathematically and why it is not the final capital.

Written record. If (u_n) is geometric with common ratio q, $u_n = u_0 q^n$ ou $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

Example 6. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.5 · essential.

1. The sequence (u_n) satisfies $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write u_n then calculate u_{12}.
2. A quantity is worth 500 and decreases by 2% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, course notes pp. 25–28; exercises p. 25, no. 19–25, then p. 28, no. 42–44. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A reforestation association plants 1,500 trees in the first year of its programme. Thanks to new volunteers, the number of trees planted doubles every year. Let $u_0 = 1{,}500$ be the number of trees planted in the first year and u_n the number of trees planted in year $(n+1)$.

a. Justify that (u_n) is a geometric sequence. Express u_n in terms of n and deduce u_6.

b. Calculate $S = u_0 + u_1 + \cdots + u_6$ and interpret this result in context.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 145 and the rate or common difference/ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 6 — Review and formative assessment

Number Sequences

1 Oct 2026 • B: 1 h – C: 2 h

Objectives.

- Choose the right model.
- Use the formulas.
- Analyse one's own mistakes.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 180. .....
2. How many terms from u_3 to u_{15}? .....
3. Give the multiplier for a 8% decrease. .....
4. Give the common difference of $12, 12+9, 12+2\times 9, \ldots$. .....
5. Calculate u_4 if $u_0 = 19$ and $u_{n+1} = u_n + 3$. .....

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: choose the right model and use the formulas.

Activity 6. Sort eight situations into three categories: arithmetic, geometric, neither. For each situation, underline the word or piece of data that justifies the choice, then choose the appropriate formula.

Written record. Before calculating, identify: type of sequence, first index, common difference or ratio, and the term or sum asked for. The conclusion states the unit and the index.

Example 7. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Exercise 1.6 · essential.

1. The sequence (u_n) satisfies $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is worth 520 and decreases by 3% per period. Write v_n and calculate v_5.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

Further practice: Calao, p. 29, no. 49–52. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. For each statement, say whether it is true or false. Justify your answer.

1. The price of an annual travel pass goes up by €1.50 every year. This situation can be modelled by an arithmetic sequence with common difference 1.5.
2. A car loses 15% of its value every year. This situation can be modelled by a geometric sequence with common ratio 1.15.
3. A sum is invested in a savings account with compound interest at an annual rate of 3%, with no deposits or withdrawals. The capital can be modelled by a geometric sequence with common ratio 1.03.
4. A rainwater tank holds 1,000 L. Each day it loses 5% of its contents, then 40 L are added. The volume can be modelled by a geometric sequence with common ratio 0.95.
5. The population of a town grows by 2% per year. This situation can be modelled by an arithmetic sequence with common difference 2.
6. A loan of €2,000 is granted with simple interest at an annual rate of 6%. The amount owed can be modelled by an arithmetic sequence with common difference 120.
7. The number of monthly visits to a website decreases by 300 every month. This situation can be modelled by a geometric sequence.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 150 and the rate or common difference/ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.